Ryohei Nagahiro

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Professional Summary

Specializing in thermal transport and thermoelectrics, with expertise in material synthesis and nanoscale thermal characterization. Research focuses on engineering grain boundary thermal resistance and evaluating local thermal transport in polycrystalline materials. Passionate about advancing sustainable energy technologies and exploring nanoscale heat transport mechanisms.

Education

Ph.D., Mechanical Engineering, The University of Tokyo, Japan 2020 – 2025
Thesis: Development of Bulk Silicon Thermoelectrics through Analysis and Control of Thermal Boundary Resistance
Supervisor: Prof. Junichiro Shiomi

M.S., Mechanical Science and Technology, Gunma University, Japan 2018 – 2020
Thesis: Thermal Analysis on Aluminum Foam Formation by Radiation Heating
Supervisor: Prof. Kenji Amagai

B.S., Mechanical Science and Technology, Gunma University, Japan 2014 – 2018

Experience

Toyota Technological Institute, Assistant Professor Jul. 2026 – Present
Department of Mechanical Engineering (Mechanical Properties of Materials Laboratory)
(Research Assistant May 2025 – July 2025)
Researching thermal transport phenomena in Prof. Junichiro Shiomi's group using time- and frequency-domain thermoreflectance (TDTR/FDTR) techniques, focusing on in-plane thermal conductivity measurements of composite materials using Thermo-Optic Phase Spectroscopy (TOPS).

The University of Tokyo, Postdoctoral Researcher Aug. 2025 – Jun. 2026
(Research Assistant May 2025 – July 2025)
Researching thermal transport phenomena in Prof. Junichiro Shiomi's group using time- and frequency-domain thermoreflectance (TDTR/FDTR) techniques, focusing on in-plane thermal conductivity measurements of composite materials using Thermo-Optic Phase Spectroscopy (TOPS).

University of Illinois Urbana-Champaign, Visiting Scholar Oct. 2025
Conducted research in Prof. David G. Cahill's group on coupled mass and heat transport in cellulose nanofiber (CNF) films. Employed Immersion Thermo-Optic Phase Spectroscopy (I-TOPS) to quantify water diffusivity and in-plane thermal conductivity under controlled humidity conditions.

Northwestern University, Visiting Scholar Nov. 2022 – Dec. 2023

Conducted research in Prof. Jeffrey G. Snyder's group, in collaboration with Prof. Oluwaseyi Balogun's group, focusing on the development and application of the Gibbs Excess method to determine thermal boundary resistance in bulk polycrystalline silicon.

JSPS Research Fellowships for Young Scientists (DC2) 2021 – 2023

Studied the fabrication of bulk silicon thermoelectric materials with low thermal conductivity, aiming to preserve nanostructures that typically degrade at high temperatures. The research focused on optimizing both material and device-level performance.

University Malaysia Pahang, Study Abroad Program Oct. 2018 – Apr. 2019

Conducted research on lubricant oil film thickness measurement during minimum quantity lubrication milling, using Laser-Induced Fluorescence under the supervision of Dr. Nurrina binti Rosli.

Publications

- M. Sasaki, K. Shizume, **R. Nagahiro**, J. Shiomi and M. Goto, "Control of thermal conductivity in amorphous silicon oxide and silicon oxynitride thin films via ubiquitous transition metal doping", *Japanese Journal of Applied Physics* 65(11), 115502, (2025).
- Odufisan, A. R., Stern, B., **R. Nagahiro**, Wynnychenko, R., Chanakian, S., Shiomi, J., Isotta, E., Balogun, O. "The Frequency Domain Thermoreflectance Technique for Thermal Property Measurements", *Journal of Visualized Experiments (JoVE)* (226), e68908, (2025).
- **R. Nagahiro**, B. Xu, S. Terashima, Y. Li, Y. Li, Y. Liao, Z. Fang, C. Shao, M. Ohnishi, S. Kato, E. Iwase, J. Shiomi, "Module-scale silicon 3D softened nanoarchitectures for eco-friendly thermoelectric energy harvesting", *Materials Today Physics*, 57, 101798 (2025).
- E. Isotta, **R. Nagahiro**, A. R. Odufisan, J. Shiomi, O. Balogun, G. J. Snyder, "Thermal boundary resistance model via mean free path suppression functions and a Gibbs excess approach", *International Journal of Heat and Mass Transfer*, 252, 127417 (2025).
- E. Isotta, S. Jiang, R. Bueno Villoro, **R. Nagahiro**, K. Maeda, D. A. Mattlat, A. R. Odufisan, A. Zevalkink, J. Shiomi, S. Zhang, C. Scheu, G. J. Snyder, O. Balogun, "Heat transport at silicon grain boundaries", *Advanced Functional Materials*, 34, 2405413 (2024).
- B. Xu, Y. Liao, Z. Fang, Y. Li, R. Guo, **R. Nagahiro**, Y. Ikoma, M. Kohno, J. Shiomi, "Extremely suppressed thermal conductivity of large-scale nanocrystalline silicon through inhomogeneous internal strain engineering", *Journal of Materials Chemistry A*, 11(35), 19017-19024 (2023).
- Y. Hangai, D. Kawato, M. Ando, M. Ohashi, Y. Morisada, T. Ogura, H. Fujii, **R. Nagahiro**, K. Amagai, T. Utsunomiya, N. Yoshikawa, "Nondestructive observation of pores during press forming of aluminum foam by X-ray radiography", *Materials Characterization*, 170, 110631 (2020).
- Y. Hangai, **R. Nagahiro**, M. Ohashi, K. Amagai, T. Utsunomiya, N. Yoshikawa, "Continuous foaming of multiple aluminum foam precursor by combining conveyor and optical heating" *Materials Transactions*, 61, 1703-1706 (2020).

- Y. Hangai, K. Takahashi, **R. Nagahiro**, K. Amagai, T. Utsunomiya, N. Yoshikawa, "Fabrication of aluminum foam with complex shapes using pin screen mold and effect of arrangement of pins on its surface morphology" *Journal of Porous Materials*, 27, 347-353 (2020).

- Y. Hangai, K. Takada, H. Fujii, Y. Aoki, Y. Aihara, **R. Nagahiro**, K. Amagai, T. Utsunomiya, N. Yoshikawa, "Foaming of A1050 aluminum precursor by generated frictional heat during friction stir processing of steel plate", *The International Journal of Advanced Manufacturing Technology*, 106, 3131-3137 (2020).

- **R. Nagahiro**, N. Rosli, Y. Hangai, A. Yano, K. Amagai, "Uniform heating model analysis for the formation process of Aluminium foam by spot-type halogen lamp", *International Journal of Automotive and Mechanical Engineering*, 16, 7170-7180 (2019).

- Y. Hangai, M. Ohashi, **R. Nagahiro**, K. Amagai, T. Utsunomiya, N. Yoshikawa, "Press forming of aluminum foam during foaming of precursor", *Materials Transactions*, 60, 2464-2469 (2019).

- Y. Hangai, H. Matsushita, R. Suzuki, S. Koyama, K. Amagai, **R. Nagahiro**, T. Utsunomiya, M. Matsubara, N. Yoshikawa, "Refoaming of deformed aluminum foam by precursor foaming process", *Journal of Porous Materials*, 26, 1149-1155 (2019).

- Y. Hangai, **R. Nagahiro**, M. Ohashi, K. Amagai, T. Utsunomiya, N. Yoshikawa, "Shaping of aluminum foam during foaming of precursor using steel mesh with various opening ratios", *Metals*, 9, 223 (2019).

- Y. Hangai, K. Takahashi, **R. Nagahiro**, K. Amagai, T. Utsunomiya, "Shaping of aluminum foam using point group mold", *The Japan Institute of Metals and Materials*, 59, 1952-1955 (2018).

(A complete and regularly updated version is also available on Google Scholar.)

Conference Presentations (since 2020)

International

- **R. Nagahiro**, N. Chowdhury, J. Peng, K. Daicho, D. G. Cahill, J. Shiomi, “Transport of Water in Cellulose Nanofiber Materials: Humidity-Dependent Mass and Thermal Transport Properties”, *The 11th US-Japan Joint Seminar on Nanoscale Transport Phenomena*, P-37 (poster), Tokyo, Japan, June 2026.

- E. Isotta, **R. Nagahiro**, A. Odufisan, J. Shiomi, O. Balogun, G. J. Snyder, “A Thermal Boundary Resistance model via mean free path suppression functions and a Gibbs excess approach”, *2025 MRS Spring Meeting & Exhibit*, SF01.09.02 (poster), Seattle, Washington, USA, April 2025.

- **R. Nagahiro**, E. Isotta, G. J. Snyder, O. Balogun, J. Shiomi, “Validation of Gibbs Excess methods with bonded Si wafer”, *The 1st A3 Nano & Thermal Energy Engineering Workshop*, P1-16 (poster), Tokyo, Japan, January 2025.

- B. Xu, **R. Nagahiro**, S. Terashima, C. Shao, Y. Li, Z. Fang, Y. Li, Y. Liao, M. Ohnishi, S. Kato, E. Iwase, J. Shiomi, “Strain Engineering in 3D Architected Silicon for High-Performance Thermoelectric Energy Harvesting”, *The Third Asian Conference on Thermal Sciences*, O-0034 (oral), Shanghai, China, June 2024.

- **R. Nagahiro**, K. Maeda, E. Isotta, S. Jiang, G. J. Snyder, O. Balogun, J. Shiomi, “Direct observation of thermal boundary resistance in silicon system”, *The Third Asian Conference on Thermal Sciences*, O-0634 (oral), Shanghai, China, June 2024.
- K. Shizume, **R. Nagahiro**, M. Sasaki, M. Goto, J. Shiomi, “Machine-Learning Minimization of Amorphous Heat Conduction by High-Throughput Deposition Process in The Loop”, *2024 MRS Spring Meeting & Exhibit*, MT03.02.08 (oral), Seattle, Washington, USA, April 2024.
- **R. Nagahiro**, B. Xu, S. Terashima, C. Shao, Y. Li, Z. Fang, Y. Li, Y. Liao, M. Ohnishi, S. Kato, E. Iwase, J. Shiomi, “Thermoelectric performance of bulk silicon with 3D nanoarchitecture”, *The 39th Annual International Conference on Thermoelectrics* (oral), Seattle, Washington, USA, June 2023.

* Full list of domestic conference presentations is available on my Website.

Patents

- J. Shiomi, B. Xu, **R. Nagahiro**, ”Manufacturing Method of a Silicon Bulk Thermoelectric Conversion Material”, United States Patent, Provisional Application filed under U.S. Patent No. 63/316,571.

Last updated: August 4, 2026